



TELEM1, TELEM2 ports

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (blk)	TX (OUT)	+3.3V
3 (blk)	RX (IN)	+3.3V
4 (blk)	CTS	+3.3V
5 (blk)	RTS	+3.3V
6 (blk)	GND	GND

GPS2 port

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (blk)	TX (OUT)	+3.3V
3 (blk)	RX (IN)	+3.3V
4 (blk)	SCL I2C1	+3.3V
5 (blk)	SDA I2C1	+3.3V
6 (blk)	GND	GND

ADC

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (blk)	ADC IN	6.6Vmax, pin 8
3 (blk)	GND	GND

I2C2

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (blk)	SCL	+3.3 (pullups)
3 (blk)	SDA	+3.3 (pullups)
4 (blk)	GND	GND

CAN1&2

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (blk)	CAN_H	+12V
3 (blk)	CAN_L	+12V
4 (blk)	GND	GND

POWER1

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (red)	VCC	+5V
3 (blk)	CURRENT	up to +3.3V,pin 15
4 (blk)	VOLTAGE	up to +3.3V,pin 14
5 (blk)	GND	GND
6 (blk)	GND	GND

POWER2

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (red)	VCC	+5V
3 (blk)	CURRENT	up to +3.3V,pin 4
4 (blk)	VOLTAGE	up to +3.3V,pin 13
5 (blk)	GND	GND
6 (blk)	GND	GND

USB

Pin	Signal	Volt
1 (red)	VCC	+5V
2 (blk)	D_plus	+3.3V
3 (blk)	D_minus	+3.3V
4 (blk)	GND	GND
5 (blk)	BUZZER	battery voltage
6 (blk)	Boot/Error LED	

8 main outs - Motors (PWM) and Servos
6 aux outs - peripherals like servo (pwm) or esc (dshot)
1 RCin - PPM/Sbus

SETUP :

1. Firmware selection (ArduCopter / ArduPlane / ArduRover)
2. Firmware flash / update
3. Frame type selection (Quad / Hexa / Octa / VTOL, etc.)
4. Accelerometer calibration - only when outdoor
5. Telemetry setup
6. GPS configuration (baud, type)
7. Compass calibration (internal / external) - only when outdoor
8. Radio setup and calibration (RC endpoints, trims)
9. Flight modes mapping (Mode switch setup)
10. Failsafe setup (RC failsafe, GCS failsafe)
11. ESC calibration / protocol setup (PWM / DShot)
12. Motor order and direction verification
13. Arming checks configuration
14. Autonomy setup (Altitude, speed, descent)
15. Peripheral setup
16. Mission Upload and Path Planning
17. Status and Actions Tab

1. COM Port Selection (Initial Connection)

- Connect the **Pixhawk Flight Controller** to your laptop using a **USB cable**.
- Open **Mission Planner**.
- On the **top-right corner** of the Mission Planner window:
 - Select the correct **COM port** (it appears automatically once Pixhawk is connected).
 - Set the **baud rate** (default is usually **115200**).
- At this stage, **you do NOT need to connect** to the board yet.

2. Firmware Selection & Flashing (ArduCopter / ArduPlane / ArduRover)

Steps to Flash Firmware

1. Open **Mission Planner**.
2. Go to the **Setup** tab using the icons on the **top-left corner**.
3. Click on **Install Firmware**.
4. You will see firmware options such as:
 - ArduCopter (for Quad, Hexa, Octa, etc.)
 - ArduPlane
 - ArduRover
5. Click on the firmware you want to install

What Happens Next

- Mission Planner will prompt you to **connect the Pixhawk via USB**.
- Once connected:
 - Firmware download starts automatically.
 - Firmware is flashed onto the Pixhawk.
- After completion:
 - Mission Planner will **disconnect the board automatically**.

Important Reboot Step

- **Unplug the USB cable** from the Pixhawk.
- **Reconnect the USB cable** again to power the Pixhawk.
- This reboot ensures the **new firmware loads correctly**.
- Now you can connect normally using the COM port.

This completes the firmware flashing process.

3. Frame Type Selection (Quad / Hexa / Octa / VTOL, etc.)

Steps to Select Frame Type

1. In **Mission Planner**, go to the **Setup** tab.
2. Click on **Mandatory Hardware**.

3. Select **Frame Type**.
4. Choose the frame that matches your drone:
 - Quad
 - Hexa
 - Octa
5. Save the selection.

Once the frame is selected, Mission Planner automatically configures internal motor mappings for that frame.

4. Accelerometer Calibration (*Outdoor Only*)

Accelerometer and compass calibration must be done outdoors.

Closed labs/workshops contain metal structures, electrical wiring, and magnetic interference that can corrupt calibration results.

Steps for Accelerometer Calibration

1. Open **Mission Planner**.
2. Go to **Setup** → **Mandatory Hardware** → **Accelerometer Calibration**.
3. Click the **green “Calibrate Accelerometer” button** (top button of all the green bars).

Calibration Procedure

- Mission Planner will guide you step-by-step.
- Perform each action exactly as shown on the screen:
 - Level the vehicle
 - Place it on its **left side**
 - Place it on its **right side**
 - Place it **nose up**
 - Place it **nose down**
 - Place it **upside down**
- Hold the vehicle steady in each position until Mission Planner confirms it.
Once all orientations are completed successfully, the accelerometer calibration is done.

5. Telemetry Setup (UART / Wireless Communication)

Telemetry setup is done **early**, so further configuration can be done **wirelessly** without connecting Pixhawk to the laptop via USB.

Serial Port Mapping (Important)

- **SERIAL1** → Telemetry 1
- **SERIAL2** → Telemetry 2
- **SERIAL3** → GPS 1
- **SERIAL4** → GPS 2

These are all **UART** ports.

The port name does **not** restrict functionality—any port can be assigned any role.

Steps to Configure Telemetry

1. Go to **Setup** → **Mandatory Hardware** → **Serial Ports**.
2. Identify the port where telemetry is physically connected
(e.g., *Telemetry connected to TELEM1* → *SERIAL1*).
3. Set the following:
 - **Protocol** → **MAVLink2**
 - **Baud Rate** → **57600**
(Most telemetry radios use 57600, confirm if specified otherwise.)

Telemetry is now configured.

You can disconnect USB and connect wirelessly using the telemetry radio.

6. GPS Configuration

GPS can be connected using **UART** or **CAN**.

Both methods are explained below.

A. GPS Configuration via UART (TX–RX)

Steps

1. Go to **Setup** → **Mandatory Hardware** → **Serial Ports**.
2. Identify the serial port connected to GPS
(example: *GPS connected to GPS1* → *SERIAL3*)
3. Set:
 - **SERIAL3 Protocol** → **GPS**
 - **SERIAL3 Baud Rate** → **115200**
(Most modern GPS modules use 115200.)

Required Parameter Check

Go to **Config** → **Full Parameter List** and ensure:

- `GPS_TYPE = 1`
- `GPS_AUTO_CONFIG = 1`
- `GPS_AUTO_SWITCH = 1`

These are usually enabled by default, but must be verified.

No I2C configuration is required for UART-based GPS.

B. GPS Configuration via CAN (DroneCAN – Preferred)

CAN-based GPS is **more reliable** and is the preferred setup.

Steps

1. Connect the GPS module to the **CAN1 port**.
2. Go to **Config** → **Full Parameter List**.
3. Set the following parameters:
 - `CAN_P1_DRIVER = 1`
 - `CAN_P1_PROTOCOL = 1`
 - `GPS_TYPE = 9` (*DroneCAN*)
4. **Reboot the Pixhawk.**

Baud rate configuration is **not required** for CAN-based GPS.

7. Compass Calibration (*Outdoor Only*)

Compass calibration must be done **outdoors** to avoid magnetic interference from buildings, metal objects, and power lines.

Steps for Compass Calibration

1. Go to **Setup** → **Mandatory Hardware** → **Compass**. Then ask ISHANN !

8. Radio Setup and Calibration (RC Endpoints & Trims)

Pixhawk can receive RC inputs in **two main ways**:

A. PPM / SBUS Receiver (RCIN Port)

- PPM or SBUS receivers are connected to the **RC IN** port of Pixhawk.
- This port is physically located near **MAIN OUT / AUX OUT** pins.

Setup Notes

- **No serial port configuration is required** for RC IN.
- Radio transmitter must already be:
 - Bound to the receiver
 - Channel mapped correctly
 - Endpoints set in the transmitter

Radio Calibration Steps

1. Go to **Setup → Mandatory Hardware → Radio Calibration**
2. Confirm that stick movements are visible on the screen
3. Click **Calibrate Radio**
4. Move **all sticks and switches** to their extreme positions
5. Follow on-screen instructions
6. Save calibration

Once completed, RC setup using PPM/SBUS is finished.

B. ELRS Receiver (UART-Based RC)

ELRS receivers **do NOT use RC IN**.

They communicate digitally using **UART (CRSF protocol)**.

Serial Configuration

1. Connect ELRS receiver to **any UART port** (TX–RX)

- Example: GPS1 port → **SERIAL3**
- 2. Go to **Setup** → **Mandatory Hardware** → **Serial Ports**
- 3. Set:
 - **SERIAL3 Protocol** → **RCIN**
 - **SERIAL3 Baud Rate** → **420000** (preferred)
(115200 works but is sub-optimal)

Radio Calibration (Same as PPM)

- Go to **Radio Calibration**
- Ensure inputs are visible
- Click **Calibrate Radio**
- Move all sticks and switches fully
- Save

9. Flight Mode Mapping (Mode Switch Setup)

Step 1: Assign Flight Mode Channel

1. Go to **Config** → **Full Parameter List**

Search:

FLTMODE_CH

2. Set this to the channel connected to your mode switch
(commonly CH5)

Step 2: Assign Flight Modes

1. Go to **Setup** → **Flight Modes**
2. You will see:
 - Current PWM value
 - Active flight mode
3. Assign up to **6 flight modes** using dropdowns

Common Flight Modes

Stabilize

- Pilot has **full control**

- No altitude or position hold
- Self-levels using accelerometer
- Used for:
 - Manual flying
 - Emergency recovery
 - First test flights

Altitude Hold (AltHold)

- Holds **altitude automatically**
- Pilot controls horizontal motion
- Throttle stick controls climb/descent

Loiter

- Holds **position and altitude**
- Requires **GPS lock**
- Ideal for:
 - Hovering
 - Observation
 - Safe pause mid-flight

Auto

- Executes **preplanned mission**
- Waypoints, takeoff, landing are autonomous
- No manual stick input required
- Used for survey and mapping missions

Guided

- Controlled by **Companion Computer**
- Can command:
 - Go to point
 - Follow target
 - Precision landing
- Used for autonomous operations

Return to Launch (RTL)

- Aircraft returns to **home position**
- Uses GPS
- Climbs to RTL altitude, then returns and lands
- Primary **failsafe mode**
- Go to config - full parameter list ->
 RTL_ALT - to set at what altitude it will make RTL
 RTL_ALT_TYPE - relative to home

10. Failsafe Setup

Failsafes ensure **safe behavior during failures**. Go to **Setup** → **Mandatory Hardware - Failsafe**

A. RC Failsafe

1. Enable **Radio Failsafe**
2. Choose action:
 - **RTL** (recommended)
 - **LAND** (only for GPS-denied or indoor use)

B. GCS Failsafe (Telemetry Loss)

- Triggers when telemetry link is lost

Recommended

- Enable **GCS Failsafe**
- Set timeout (seconds) , given in next page, how to setup
- Action → **RTL**

C. Battery Failsafe (Very Important)

Battery Monitor Setup

1. Go to **Setup** → **Optional Hardware** → **Battery Monitor**
2. Set:
 - **Monitor** → **Analog Voltage and Current**

- **Sensor** → Select your power module
(Holybro / Cube / Hobbywing etc.)
3. Verify voltage & current readings

Voltage Thresholds (LiPo)

- **Failsafe (RTL)** → ~3.3 V/cell
- **Critical** → 3.2 V/cell (avoid reaching)

D. Advanced Failsafe Parameters (FS_)

Go to **Config** → **Full Parameter List**, search **FS_**

Recommended Settings

- **FS_CRASH_CHECK = 1**
→ Disarms motors on crash detection
- **FS_DR_ENABLE = LAND**
→ GPS loss → LAND (NOT RTL)
- **FS_GCS_ENABLE = 1**
- **FS_THR_ENABLE = RTL**
- **FS_GCS_TIMEOUT** → set as needed (seconds)

E. GeoFence Failsafe

1. Go to **Config** → **Full Parameter List**
2. Search **FENCE_**

Key Parameters

- **FENCE_ENABLE = 1**
- **FENCE_ACTION = RTL or LAND**
- **FENCE_ALT_MAX** → Maximum allowed altitude
- **FENCE_MARGIN** → Buffer distance before violation, when a survey mission is uploaded
If the drone exits the fence → failsafe is triggered.

11. ESC Calibration & Protocol Setup (PWM / DShot)

Pixhawk supports multiple ESC communication protocols.

The two most commonly used are **PWM** and **DSHOT**.

A. PWM ESC

PWM ESCs require **manual calibration** so the ESC learns throttle endpoints.

Steps for PWM ESC Calibration

1. Open **Mission Planner**
2. Go to **Setup** → **Mandatory Hardware** → **ESC Calibration**
3. Select **ESC Type = Normal**
4. Leave all other settings as default
5. Click **Calibrate ESC**

Calibration Procedure

1. **Disconnect USB cable**
 2. **Disconnect battery**
 3. Reconnect **battery**
 4. Wait for **several seconds**
- Motors will emit **beep tones**
 - Beeping will stop once calibration is complete

Once beeping stops, ESC calibration is done.

Important Parameter Check (Recommended)

1. Go to **Config** → **Full Parameter List**

Search:

BRD_IO_ENABLE

Ensure:

BRD_IO_ENABLE = 1

This enables the IO co-processor, providing **stable and precise PWM motor outputs**.

B. DShot ESC Setup (Digital Protocol)

DSHOT is a **digital protocol** and **does NOT** require ESC calibration.

Important Hardware Rule

- DShot ESCs must be connected to AUX outputs
- PWM ESCs must be connected to MAIN outputs

DSHOT Configuration Steps

1. Go to **Config** → **Full Parameter List**

Search:

BRD_IO_DSHOT - enable

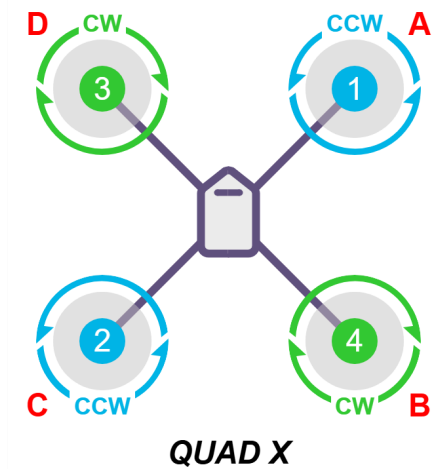
2. Select the appropriate firmware: Go to **Setup** → **Mandatory Hardware** → **ESC Calibration**
3. Select **ESC Type** -
 - **DShot300**
 - **DShot600** (*most commonly used*)
4. Reboot the Pixhawk

Motor Output Mapping (AUX Pins), setup - mandatory hardware - servo output :

- **Servo 9** → AUX 1 → Motor 1
- **Servo 10** → AUX 2 → Motor 2
- **Servo 11** → AUX 3 → Motor 3
- **Servo 12** → AUX 4 → Motor 4

Connect motors accordingly to AUX pins. Since DShot is digital, **no throttle calibration is required**.

12. Motor Order & Direction Verification



Correct motor order and direction are **critical for flight stability**.

Motor Test Procedure

1. Go to **Setup** → **Optional Hardware** → **Motor Test**
2. Select **Throttle Percentage** ≈ 15–20%
3. Increase **Spin Duration** if needed
4. Test motors **one by one**

What to Verify

- Correct **motor number**
- Correct **motor position**
- Correct **rotation direction**
 - Clockwise (CW)
 - Counter-clockwise (CCW)

If a motor:

- Spins in the wrong direction → **swap any two motor wires** (for PWM/DSHOT ESCs)
- Spins in wrong position → **rewire to correct output**

Repeat until **all motors match the diagram**.

13. Arming Checks Configuration

Arming checks ensure that the vehicle **arms only when all critical conditions are satisfied**. If any selected check fails, Pixhawk will **refuse to arm**.

A. Arming Check Bitmask

Steps

1. Go to **Config** → **Full Parameter List**

Search:

ARMING_CHECK

2. Select only the checks that are **actually being used**, such as:
 - Gyroscope
 - Accelerometer
 - Compass
 - GPS
 - Battery

Do **not** enable checks for sensors or features you are not using, as this may block arming unnecessarily.

B. Disable Rudder Arming (Recommended)

Rudder arming is inconvenient and unsafe for multirotors.

In **Full Parameter List**, search:

ARMING_RUDDER = 0

C. Assign Dedicated Arming Switch (RC Channel)

Using a **dedicated switch** for arming is safer and more reliable.

Steps

1. Choose a free RC channel (e.g. **RC7 or RC8**)

In **Full Parameter List**, search:

RC7_OPTION = Arm / Disarm

2. *(Shown as "ARM/DISARM" in ArduPilot 4.2 and now this RC switch will be used exclusively for arming and disarming.*

D. Disable Safety Switch (If Not Used)

If no physical safety switch is installed, disable it to avoid arming issues.

Search:

`BRD_SAFETY_DEFLT = 0`

14. Autonomy Setup (Altitude, Speed, Descent)

Autonomous flight behavior is controlled primarily using **WP_NAV** parameters. These parameters define how the drone moves **between waypoints**.

Key WP_NAV Parameters

Go to **Config** → **Full Parameter List**, search `WPNAV_`

1. `WPNAV_SPEED`

Horizontal cruising speed during autonomous flight

- Units: **cm/s**

Example:

`500 → 5 m/s`

Controls how fast the drone moves between waypoints.

2. `WPNAV_SPEED_UP`

Maximum climb speed

- Units: **cm/s**
- Limits how aggressively the drone climbs

Too high → motor saturation

Too low → sluggish climb

3. `WPNAV_SPEED_DN`

Maximum descent speed

- Units: **cm/s**
- Should always be **lower than climb speed**

This prevents:

- Prop wash instability
- Vortex ring state

4. **WPNAV_ACCEL**

Horizontal acceleration limit

- Units: **cm/s²**
- Controls how aggressively the drone accelerates and decelerates

Higher values:

- Faster respons
- More stress on frame and motors

Lower values:

- Smoother, safer motion

5. **WPNAV_JERK**

Rate of change of acceleration

- Units: **cm/s³**
- Controls smoothness of motion

Lower jerk:

- Smoother paths
- Less oscillation

Higher jerk:

- Sharper turns
- Faster response

6. WPNAV_RADIUS

Waypoint acceptance radius

- Units: **cm**
- Distance from waypoint at which it is considered “reached”

Smaller radius:

- Higher accuracy
- More aggressive braking

Larger radius:

- Smoother missions
- Faster transitions

7. WP_YAW_BEHAVIOR

WP_YAW_BEHAVIOR defines **how the drone controls its yaw (heading)** during autonomous missions.

- **0 – Hold yaw**
Drone maintains the same heading throughout the mission.
- **1 – Face next waypoint (default)**
Drone automatically turns to face the direction of travel.
- **2 – Face ROI (Region of Interest)**
Drone continuously faces a specified target point.
- **3 – RC control**
Pilot manually controls yaw during AUTO mode.

This parameter is used to control **camera orientation, mission smoothness, and heading behavior** in autonomous flights

How WP_NAV Values Are Chosen

WP_NAV values **cannot be fixed universally**.

They depend on:

- Drone size
- All-up weight
- Propeller size
- Battery voltage
- Motor KV and model
- Mission type (survey / delivery / rescue)

Final values are determined through:

- Engineering judgment
- Flight testing
- Calculations (covered separately)
- AI-assisted tuning

These calculations will be taught later and should not be guessed blindly.

15. Peripheral Setup (Payload & LiDAR)

1. Payload (Servo-Based)

Payload mechanisms are typically controlled using a **servo output**.

Servo Configuration

1. Go to **Setup** → **Servo Output**
2. Select any **MAIN** or **AUX** output pin where the servo is connected.

Control Methods

- **Mission Planner / Companion Computer control**
 - Keep the servo **Function = Disabled**
 - The servo can still be triggered via:
 - Mission commands
 - Companion computer (MAVLink)

- **RC Remote control**
 - Select the servo output pin (e.g. **Servo 10** → **AUX 2**)
 - Set **Function = RCINx** (RCIN7 / RCIN8 / RCIN9, etc.)
 - That RC channel will now directly control the servo

The same RC channel can be assigned to multiple servo outputs if required.

2. LiDAR (Rangefinder Setup)

We commonly use **Benewake TF02 LiDAR**, connected via **UART**.

Serial Port Setup

1. Go to **Setup** → **Mandatory Hardware** → **Serial Ports**
2. Select the port where LiDAR is connected
(e.g. *Telemetry 2* → *SERIAL2*)
3. Set:
 - **Protocol = Rangefinder**
 - **Baud Rate = 115200** (as per TF02 datasheet)
4. Save parameters

Rangefinder Parameters

1. Go to **Config** → **Full Parameter List**

Search:

RNGFND

2. Set:
 - **RNGFND_TYPE = Benewake TF02**
 - **RNGFND_MIN_CM** = minimum valid distance
 - **RNGFND_MAX_CM** = maximum valid distance
 - **RNGFND_ORIENT = Downward Facing** (most common)
3. **Reboot the flight controller**

16. Mission Upload and Path Planning

Mission planning is done using the Mission **Planner** → **Plan** tab.

A telemetry connection is required to **load the map and set home location correctly**.

1. Setting Home Location

1. Go to the **Plan** tab.
2. Once connected, the map will load.
3. Click on the **Home Location** shown at the bottom-right corner.
4. This displays:
 - Latitude
 - Longitude
 - Home altitude (AMSL)

Home position is used as the reference point for **Relative Altitude** missions.

2. General Waypoint Mission

Creating Waypoints

1. Click anywhere on the map to create **Waypoint 1**
2. Click multiple locations to create successive waypoints
3. For each waypoint:
 - Command → **Waypoint**
 - Frame → **Relative**
 - Set **altitude** relative to takeoff point

Mandatory Mission Commands

A mission **must not consist of only waypoints**.

Add Takeoff

1. Click **Add Below** (green button)
2. Change command to **Takeoff**
3. Set **takeoff altitude**

Add End Command

At the end of the mission:

- Use **RTL (Return to Launch)**
OR

- Use **Land**

For RTL, **altitude is not set here** — it is controlled by RTL parameters.
Takeoff altitude **must always be specified**.

3. Survey Mission (Grid Mapping)

Survey missions can be created using **polygon-based planning**.

Creating or Importing a Polygon

You can:

- Draw a polygon using the **polygon icon** on the map
- OR**
- Import a polygon file

Important

- Mission Planner does **not accept KML directly**
- Polygon must be imported in a compatible format
(*conversion can be done externally if required, google earth engine*)

Generating Survey Grid

1. Right-click on the polygon
2. Select **Auto Waypoints** → **Survey (Grid)**

Survey Settings (Minimal & Manual Method)

Simple Tab

- Set **Flight Altitude**
- Tick:
 - ☒ Add Takeoff and Land Waypoints
 - ☒ Use RTL
- Ignore camera angle (manual calculation used)

Grid Options Tab

- Set **Distance Between Lines**
(*calculated manually based on camera & mission requirements*)

All other parameters can be left default.

Mission Preview

At the bottom, Mission Planner shows:

- Survey area covered
- Estimated flight time

Camera trigger options may appear —
ignore them

4. Uploading the Mission

1. Review:
 - Waypoint order
 - Altitude values
 - Frame type (Relative)
2. Click **Write WPs**

Once written:

- Mission is uploaded to the flight controller
- Takeoff, survey path, and RTL/Land will be visible

Starting an Autonomous Mission (AUTO Mode)

1. **Arm the vehicle in a manual flight mode**
(*Stabilize / AltHold / Loiter*)
Auto and Guided modes are not armable.
2. Once armed, **switch the flight mode to AUTO** using the RC mode switch.
3. If the mission includes a **Takeoff command**, the vehicle will:
 - Automatically take off
 - Start following the mission waypoints
(*Throttle input is ignored in AUTO mode*)
 -

4. During the mission, **switching out of AUTO** to:

- Stabilize
- AltHold
- Loiter

Immediately returns **full manual control** to the pilot.

18. Data Tab:



Quick Tab

- Shows **live HUD (artificial horizon)** and **basic telemetry**
- Used for:
 - Visualizing attitude (roll, pitch, yaw)
 - Seeing arm/disarm state
 - Checking pre-arm messages on screen
- **No commands are sent from here**
 - It is mainly for **quick situational awareness**

Actions Tab

The **Actions tab** is used to **send immediate commands** to the flight controller.

Common buttons you see here:

- **Arm / Disarm** → Arms or disarms the vehicle
- **Set Mode** → Changes flight mode (Stabilize, Auto, RTL, etc.)
- **Takeoff / Auto / RTL / Loiter** → Commands the vehicle directly
- **Restart Mission / Resume Mission** → Controls mission execution
- **Abort Landing** → Cancels an ongoing landing
- **Set Home Alt / Set Loiter Rad / Change Speed / Change Alt** → Modifies mission parameters in flight

Used mainly for **guided flight, testing, and emergency actions**.

Messages Tab

The **Messages tab** shows **all system messages and warnings**.

What you see here:

- **PreArm warnings**
 - (e.g. *Compass not calibrated, RC not found*)
- **Sensor errors**
- **EKF and GPS issues**
- **Failsafe triggers**
- **System status messages**

If arming is blocked, **this tab tells you exactly why**.

Status Tab

The **Status tab** shows **raw live telemetry values** from the flight controller.

Typical values include:

- Attitude data (roll, pitch, yaw)
- GPS position (lat, long, alt)
- Battery voltages (cell-wise)
- Armed / Disarmed state
- Altitude, airspeed, acceleration
- EKF and sensor outputs
- Range Finder Values

Used for **debugging, verification, and detailed monitoring**.